Hybrid Neural Networks based Approach for Holoscopic Micro-Gesture Recognition in Images and Videos

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***Abstract*—This paper presents an approach for hand based micro-gesture recognition in images and videos as part of the Holoscopic Micro-Gesture Recognition (HoMGR) challenge. The database consists of Holoscopic 3D Micro-Gesture images and videos. The proposed framework is an ensemble of convolutional neural network and deep neural network. The framework performs feature fusion technique on both handcrafted (local phase quantization) and deep features extracted from the neural network, to leverage on complimentary information. The powerful discriminative nature of the fused features has proved beneficial on the given HoMGR challenge data. The experiments show that the proposed approach is effective and outperforms the baseline on the Test set by an absolute margin of 26.67% for images and 2.47% for videos, respectively.**

***Keywords*-Holoscopy; Micro gesture recognition**

I. Introduction

With the advancement in technology, human beings have experienced many changes in various walks of life. Advancement in technology aims to ease the way of doing activities on a day to day work. Human-Computer Interaction (HCI) is a vital research area to understand human-computer interaction pattern. This is useful to understand cognitive behaviors of human beings while interacting with machines. It has applications in Computer Vision, entertainment like watching TV, user engagement such as e-learning environment and the aim is to develop interaction of humans and computers more comfortable and interactive [1].

3D display techniques has achieved a new direction with the development of virtual reality and augmented reality. Various techniques have been introduced [2] at a different time to improve the experience of 3D display and do away with the limitations of the previous ones. Multiview stereoscopic imaging system has introduced in literature to overcome the limitations of stereoscopic imaging which required special glasses to fuse left and right view. Multiview system relies on viewing position, resolution and depends on the brain to fuse images which cause fatigue and require a prolonged focus on the screen. Later, Holograms technology [1] has introduced in the field to store refracted light from the objects as a 3D representation, but it requires the interference of only coherent light field for recording. Hence, the wide range of applications of 3D display technique requires complex hardware to achieve their objective. To overcome the limitations of the present 3D imaging display technologies, the holoscopic imaging technique is introduced.

The Holoscopic imaging technique is a method to create 3D virtual images which are based on the principle of fly's eye [3]. Such images store angular and spatial information of a scene [4]. A 3D scene can be reconstructed at different depth levels by holoscopic images. 3D information can be stored on a 2D medium which can be used to create the complete 3D model. Also, unlike Holography, coherent light sources are not required for the creation of holoscopic images.

Holoscopic 3D images are constructed by placing a number of microlenses in the form of a 2D array. Each micro lens captures the object by different viewing angle, termed as an Elemental Image (EI). The EI's are captured by a microlens, hence it is very complex to process them due to their low resolution. To overcome this, these EI's are converted to a number of Viewpoint Images (VPI) by combining each pixel present at the same position in different EI's. The main difference between VPI and the simple 2D image is that VPI is represented by parallel projections of 3D space whereas the 2D image is represented by perspective projections [5]. In Holoscopic images, pixels present at a same local position in different microlenses contain the recording of the object from the same direction [5].

Several applications of 3D images require a certain level of interaction of humans with 3D images. Hand gestures are an efficient mode available to perform all the necessary actions on 3D objects. For hand gesture recognition, data gloves were used which senses the position of gesture of the hand. However, this had a limitation of using specialized hardware for gesture recognition which doesn't make it feasible to use it in 3D display techniques. Hand gestures can be static or dynamic [6]. In static hand gestures, a fixed hand position needs to be recognized to give some input. Dynamic hand gestures represent hand pose as well as some temporal information with it. The process of hand gesture recognition can be accomplished in three phases [7]: detection, tracking, and recognition.

In this paper, we accomplished the task of hand gesture recognition in holoscopic images and videos by using deep learning methods. We classify the holoscopic images without extracting them as they contain much noise which affects the

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resulting image. Convolutional Neural Networks (CNN) are used as these are found to perform well in many appearance based classification tasks [8]. To take the advantage of different approaches, we ensemble the deep learned features and handcrafted features. Various techniques like cross-validation and data augmentation are used to train the model in a better way and prevent it from overfitting. In this paper, the only static hand gesture is recognized as we want to classify hand gestures into three classes only.

The remaining paper structure is as follows: Section II describes the concept of holoscopic images and gesture recognition and related work for the same. Section III explains the proposed method for recognizing micro-gestures in both, images and videos. Section IV and V describes the experimental setup used to implement the proposed method and its result for images and videos. Section VI discusses the conclusion and the future scope of the study.

II. Literature Review

The main idea of the Holoscopic image is taken from the concept of integral imaging [9]. Both techniques use a number of microlenses to store images but the recent advancement in the technique of reconstruction and non-dependence of special hardware (like glasses) in order to view the 3D image; is marking it as a future of 3D imaging technique.

Holoscopic images use the concept of stereo geometry. These are created by placing a number of lenses in form of a 2D array [10]. Each lens captures the image at its respective angle which is different from the other lenses. To process such image, Viewpoint Images (VPI) are extracted by selecting a pixel from each microlens. Such VPI's represent an image in orthographic projections.

The extraction of an image from a particular angle of a holoscopic 3D image is a challenging task. A wide range of methods is proposed to reduce the error encountered while it's processing. The process of capturing 3D image requires a number of microlenses placed together. Any small amount of space between micro-lenses can introduce dark borders in the holoscopic image which results in the presence of error while processing it. Kaufmann et al. [11] reduced such error by implementing a processing filter. This is done by capturing white background and doing calibration to reduce the size of dark borders. [10] provided a method which calculates the angle of deviation to identify the presence of any geometric distortion while capturing the image. The extraction of VPI's by selecting one pixel from each elemental image yields a low-resolution image. Part of the spatial information is lost in this procedure. [4] proposed a method to improve the quality of extracted images by finding a spatial correlation between the images captured at a different angle.

To increase the applicability of holoscopic images, a large amount of work has been done to reconstruct the object from 3D image after estimating its depth. [3] and [12] provided different approaches to estimate the depth of an object. [13] has done this process by using feature match selection algorithm. Zarpalas et al. [14] perform the same by calculating disparity with the help of reference points also known as anchor points.

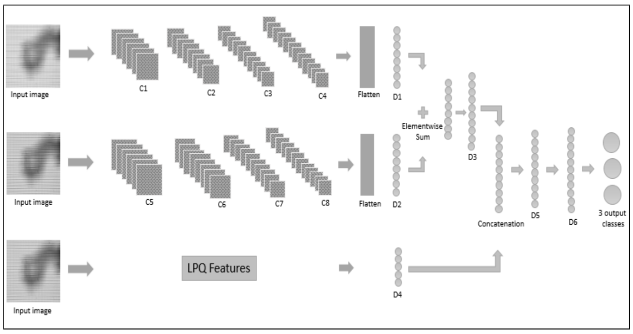
Hand gestures are the preferred modality for interacting with the constructed 3D image. Hand gestures can be used to provide spatial as well as temporal information. [15] proposed a method to detect two hand gestures: front and back in a stereo camera using color-based image segmentation. The method used in this approach includes finding the change of average pixel values in the disparity map after performing all the pre-processing tasks. The learning was performed using a conditional random field algorithm. Hand gestures can also be recognized by finding depth map of the hand image. [16] detected different hand gestures with an overall accuracy of 94.7%. Different appearance based features were extracted including a number of stretched fingers and the angle between the fingers. Later on, decision tree method was used to recognize the hand gesture.

Apostol et al. [17] also proposed a similar process to recognize static hand gestures in videos. Local spin image descriptor was used on 3D point cloud formed from the depth map of the image. SVM classifier was used for learning with 72.75% of overall accuracy. In [18] a method is proposed to recognize hand gestures on integral images. The sequence of 3D video scenes was used to detect three static hand gestures. The process was done by detecting spatio temporal interest points for each video and using a bag of words for classification using interest points and its created vocabulary. 88.98% of accuracy on an average was achieved by this process.

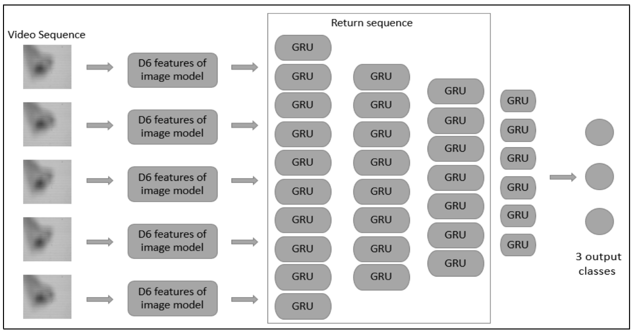
Liu et al. [19] propose a new database for gesture recognition from holoscopic images. To the best of our knowledge, this is the first ever dataset of holoscopic images collected for the micro-gesture recognition task. Gesture recognition was performed for both images and videos separately for the first time for holoscopic images. For videos, Local Binary Pattern in Three Orthogonal Planes (LBP-TOP) [20] like descriptors were computed. For the gesture recognition task, three classifiers: k-nearest neighbor, support vector machines, and naive bayes classifiers are compared. For the image based classification task, Local Binary Pattern (LBP) [21] and Local Phase Quantization (LPQ) [22] features were evaluated along with the same three classifiers. The classification was done without using any 3D pre-processing task for images and videos. The overall accuracy of 50.9% for images and 66.7% for videos was achieved by the given approach.

III. Proposed Network

In this section, we discuss the detailed architecture of our proposed network. Along with the CNN, we have also



(a) Network used for image-based gesture recognition.



(b) Network used for video-based gesture recognition.

Figure 1: Overview of the networks. (a) Image gestures are recognized using handcrafted and deep learned features. LPQ refers to the handcrafted features and CNN (C), DNN (D) are used for deep-learning features. (b) Video-based gestures are recognized using deep learned features of image network. For learning temporal information of the sequence, we used several layers of GRU.

computed the handcrafted LPQ features. The high-level features extracted using CNN show improvement in the term of learning and achieve the state-of-the-art results in various appearance based challenges [8]. CNN is well known for learning the edges and the pattern as we go deeper in the network. On contrary, the LPQ features are known to be robust against the blur and illumination changes [22]. Motivated by the use of LPQ in the baseline [19], we use it as the hand crafted feature descriptor in our network. In the experiments, ensemble of CNN (based on different filter sizes) are also evaluated. The details of the framework are discussed below.

A. Image Based Gesture Recognition

We first discuss the pipeline for the image based micro-gesture recognition. Gesture Recognition is a task where the network must be able to classify the image in a particular class using the relative information among the fingertips such as the gap between them and the posture of the fingers as a whole. Thus a large filter size was a good option to tackle the challenge. The architecture can be seen in Figure 1a. In our proposed network, we applied two CNN's namely CNN_{K-15} and CNN_{K-21}. The naming convention is based on the input filter size. The choice of the filter size is based on our observation that the CNN with small filter sizes (3×3 and 5×5) were unable to learn the task. Both the networks with larger filter sizes (15×15 and 21×21, respectively) performed better, individually. When they are fused together better performance was observed.

First, the architecture of CNN_{K-15} is explained. The convolution with 64 filters of size 15×15 is applied as C1. It is then followed by C2 with 128 filters of size 7×7, C3 with 192 filters of size 3×3 and C4 with 256 filters of size 3×3. We include the max pooling layer with filter size 3×3 after C1, C2 and C4. Once the image is passed through all the convolutional layers, flatten layer is applied. Finally, the best 4096 dimension features of CNN_{K-15} are obtained by applying a dense layer D1.

In parallel, the architecture of CNN_{K-21} consists of convolutional layers: C5 with 64 filters of size 21×21, C6 with 128 filters of size 13×13, C7 with 192 filters of size 5×5 and C8 with 256 filters of size 3×3. Here again, we include max pooling layer with filter size 3×3 after C5, C6 and C8. 4096 dimension features of CNN_{K-21} are obtained by applying dense layer D2 in succession to the flatten layer. It is found that element-wise sum operation on the best features of each network is found to be performing better than a direct concatenation of the features. Element-wise sum operation gives 4096 features. These features are then forwarded to the dense layer D3 with 4160 nodes.

It is observed that images in the dataset, are little blurred. Also, it is noted in the baseline paper that the LPQ features performed well in computing the baseline results. Motivated by this, we input the LPQ features to the dense layer D4 with 256 hidden nodes and later these 256 features were concatenated to the so far obtained 4160 features. These features are now used to predict the final class after iterating through D5 and D6 dense layers with 8448 and 8704 nodes, respectively. To avoid over-fitting we used 0.4 and 0.2 as the dropout measures after D5 and D6, respectively. After obtaining the final class label, we evaluated several cross-validation and voting strategies, which are discussed in Section V.

1) Training Network: Training of a network can be highly affected by freezing the modification of the weights in intermediate layers during back-propagation. In the case of a merge, each network can be individually trained on the final labels to obtain the best intermediate features. Later on, the best features can be merged and again the training can be done with final labels. Here, we applied joint-fine tuning [23]. We allow the network to back-propagate to the input image and LPQ features, and thus can alter the values of the weights in every layer. Joint fine tuning helps in modifying the network as a whole rather than improving each individual network.

B. Video Based Gesture Recognition

For the task of micro-gesture recognition in videos, we use the frame level information as an input image for the network. Our final prediction for video label depends mainly on the temporal pooling. Firstly, we divide all the videos in a same number of frames. This is described in detail in Section V. Each frame of a video sample is passed as input to the image network (discussed above). The information from the second last dense layer D6 was extracted.

Each sequence of images is now converted to the sequence of the features. Dealing with the sequence, Gated Recurrent Unit (GRU) [24] and Long Short-Term Memory (LSTM) are the most effective techniques for deep learning. Here, we use GRU as the sequence length (which we kept as 45 frames per video) was not sufficient to utilize the effectiveness of LSTM. Each sequence is forwarded to the 4 layers of GRU. The first three layers are allowed to return the output as a sequence so that the flow of data to the next GRU layer can be easily possible. Starting from the 256 GRU's in the first layer and then forwarding the output sequence to the further layers containing 128 GRU's, 64 GRU's and finally 32 GRU's, the sequence information is mapped to 32 nodes. After obtaining these 32 features, a dense layer with 3 nodes and activation function as softmax, is applied to obtain the final class prediction. The cross-validation and voting strategies are implemented for video gesture recognition as well. Figure 1b depicts the proposed architecture for video gesture recognition. In the next section, we are going to discuss the training of our video-based network.

1) Training Network: For video gesture recognition, we first tried to apply joint fine-tuning and allowed the weights to be updated till the first layer i.e image input during learning. The results were not convincing w.r.t the large training error. We argue that the reason for the large error in the joint fine tuning is may be due to the depth of the network. Joint fine tuning asks for the learning of a really large number of parameters and thus can be a bit complex in terms of learning (for a simpler task). We chose to freeze the network till the D5 layer of the image model and only allow an update to the weights of the GRU layers. The network now is found to be learning well.

IV. Experimental Setup

A. Image Based Gesture Recognition

We use the 3D Micro-Gesture (HoMG) database [19] which is the only such database available, to evaluate the proposed network for image gesture recognition. The dataset contains three classes for micro-gestures: *button*, *dial* and *slider*. These micro-gestures are selected with the basic idea of controlling the display in 3D images. Button to be used for submission, dial to change the current situation and slider to move window or object up and down. The distance between the object and the lenses varies, some were shot from very close distance and some from far.

The database is entirely person independent i.e. the subjects are mutually exclusive among the *Train*, *Validation* and the *Test* sets. There are 30614 images in total in the database. The *Train* set has 16763 images, which further are divided into 8364 images for close and 6853 images for far distance capture. Note that this meta-data information is not available. The *Validation* set has 6560 images (3077 images in close and 3879 images in the far distance). The *Test* set contains 7291 images among which 3930 images are in close and 4512 images are in the far distance.

1) Data processing: The data processing is required as the raw images are of 1080×1920 pixels. Dealing with this high-resolution images requires a very high computation power. We attempted to perform the experiment on resizing the images to 36×64, 54×96 and 128×128. The results on 128×128 were observed to be the best and thus all the later on mentioned experiments are performed with an input image of size 128×128.

2) Data Augmentation: The data augmentation is applied to increase the variation of the data. Each image is flipped horizontally as the dataset contains the hand images of the person standing to the right as well as of a person standing to the left of the camera. By visualizing the images, it is also found that to detect these three micro-gestures, only spatial locations of the fingers of the hand are required. The hand is located in the center for all the images. The dataset contains the images shot from variable distance and thus the scaling of the image with different scaling factors could be beneficial for data augmentation. As we chose the scaling to be done with two scaling factors, we obtained three images (two scaled and one flipped) by each image.

3) Data Splitting: In order to use both *Train* and *Validation* dataset for training, we use k-fold cross-validation to evaluate our model. Cross-validation divides the dataset to ensure unbiased results and to prevent overfitting. Cross-validation is done such that augmented data is used only for training and not for validation. Data is divided randomly for training and validation for cross-validation. We applied 3-fold and 4-fold cross validation for obtaining the best results. After applying cross-validation, the voting technique is used for the final prediction of the images using each trained model of the cross-validation.

B. Video Based Gesture Recognition

We used the videos from the 3D Micro-Gesture (HoMG) database [19] to evaluate our proposed network. The dataset is similar to the image dataset referred in section IV-A. The video subset consists of videos recorded by 40 subjects with each subject having 24 videos. Video subset is divided into 20 subjects for training and 10 for development and testing. All the videos are recorded at 25 fps with temporal length ranging from 2 to 20 seconds.

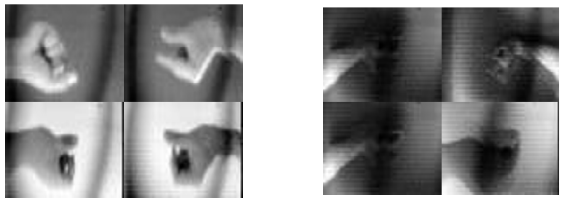
1) Data processing: To process video dataset, a fixed number of frames need to be selected. By visualizing the dataset, it is noticed that mostly the video at center contains enough information for the prediction of the label and hence, we try to crop the frames from the center of the video for both training and testing. The performance is measured by taking 31 and 45 frames from each video. The selection of 45 frames is found to be a better choice and thus, we performed all the experiments by selecting 45 frames. Each frame is later on resized to 128×128 pixels to utilize the benefit of our image network.

2) Data Augmentation: To perform the data augmentation in videos, each video is divided into a different number of segments containing 45 frames each. This data augmentation is performed such that, division starts from the center of the video i.e. the centermost 45 frames contribute to the formation of the first segment. Later 45 frames prior to the center 45 are marked as the second segment and the 45 frames after the first segment (center) are marked as the third segment, and so on. This is done to ensure that each video is included at least once for the training. The first segment from each video is treated as the main segment. To ensure the performance of the *Test* set, the prediction is done only on the main segment of each test video. Experimentally, it is observed that the segment length (45 frames) with data augmentation results in the increase of the performance.

3) Data Splitting: For the evaluation on the *Test* set in the video based micro-gesture recognition, we use both *Train* and *Validation* sets to train our model. We used 4-fold, 6-fold and 9-fold cross-validation. This was done to utilize the *Train* and *Validation* data for training and also to prevent the model from overfitting. The cross-validation is done only on the main segments and later on, for every split, the augmented data for each training set has been added. After performing the cross-validation, the voting technique is used to predict the final labels.

C. Processing Holoscopic Images

This section discusses different techniques, we implemented to convert the Holoscopic images to View Point Images (VPI). These techniques require handling constraints such as threshold values, dealing with interest points and several assumptions over the formation of an Elemental Image (EI). These constraints lead to noisy outputs of the VPI's and hence we ignored them for further use. Later on, we discuss the results we achieved over image-based and video-based gesture recognition. It is to be noted here that the networks above and all the results are from holoscopic images and not VPI.



(a) Correct conversions to VPI. (b) Noisy conversions to VPI.

Figure 2: Sample output VPI, which are extracted from the Holoscopic images. Note that the VPI are not used in the experiments.

It is observed that in the HoMG database, few of the holoscopic images have large non-uniform dark borders. These dark borders may have been introduced during the image capturing process, as discussed in Section II. Holoscopic images are easy to compare by finding the difference in their VPI's as the EI's are of low resolution. We tried to extract VPI's from the given holoscopic images by the method discussed in [3], [13], and [25]. The extracted VPI's can be seen in Figure 2. All the pixels present at a same local position in different microlenses are combined to form VPI's. VPI represent the object from a particular view. Several operations like interest point detection, are applied to get the information of a particular EI as the pattern of dark borders are not uniform in the images.

The resulting extracted VPI's were noisy. Noise removal processes were applied and these were successful to remove the noise from the images, which were having an object at a close distance as compared to images with object at far point. The inconsistency and the variable distance between object and lens lead to the uneven formation of the generated VPI. Figure 2a refers to some correct transformations and Figure 2b refers to some noisy transformations. Due to irregularity, or the presence of many noisy VPI's we decided to ignore these images for the purpose of the experiments. This processing was an attempt to obtain the smooth VPI's but was found complex and noisy. To obtain the best results, we performed all the experiments on the holoscopic images itself ignoring all the generated VPI.

V. Experimental Results

This section emphasis on the performance of our approach w.r.t. some previously implemented approaches. Our results show an increase in the classification accuracy for the prediction of the gesture based on holoscopic images.

A. Image Based Micro-Gesture Recognition

Table I shows the results and the performance achieved by each extracted deep features (CNN features), the handcrafted features (LPQ) and their fusion, individually. The classification accuracy is computed on the *Validation* dataset and the learning is done only on *Train* dataset. From the results in the Table I, it is evident that fusing the different networks helps in capturing complimentary information. CNN_{K-15} is found to be the best model among all the individual networks. The ensemble of CNN_{K-15} and CNN_{K-21} shows an absolute increase of 1.26%. LPQ features with DNN were able to classify the labels with 37.91% only, but when merged to form our proposed network, 2.96% improvement is observed. The table also shows the benefit of using Swish [26] over ReLU as the activation function.

Table I: *Validation* acc. (%) obtained by different networks. The comparison of ReLU and Swish is shown too. (Here, CNN_{K-15} refers to the CNN with filter size 15×15, CNN_{K-21} is CNN with filter size 21×21, + indicates element wise sum operation.)

Method		*Validation* Acc. (%)
ReLU	CNN_{K-15}	71.69
	CNN_{K-21}	70.82
	$CNN_{K-15} + CNN_{K-21}$	72.95
	DNN_{LPQ}	37.91
	Fused network	74.63
Swish	CNN_{K-15}	72.72
	CNN_{K-21}	71.40
	$CNN_{K-15} + CNN_{K-21}$	73.79
	DNN_{LPQ}	41.50
	Fused Network	75.76
Baseline Paper	k-NN	32.90
	SVM	51.60
	Bayes	50.60

Proposed approach outperforms the baseline results on *Validation* dataset of images by 24.16%.

Once the model, based on the proposed network is found to be performing well in terms of its classification accuracy, we also evaluated it with other metrics. Table II shows the confusion matrix on the *Validation* dataset. Table III displays the precision, recall and F1 score.

Further, in order to generalize the model (for the *Test* evaluation), we experimented with learning on *Train* and *Validation*, together. To tune the model, different techniques of cross-validation and voting are used. The results on the *Test* set can be found in Table IV. Each trained model for every split is used for the prediction of labels for the *Test* dataset. Each *Test* set image is now having the same number of labels as the number of splits applied. Finally, the class label with the maximum frequency of prediction with different models is assigned as the final predicted label for the image. This technique is referred to as *max voting*. Top-1 refers to the prediction made by the best model in terms of split validation accuracy. The best accuracy on the *Test* set is found to be 77.57% by applying 3-fold cross-validation. We also tried to use the processed data of videos (mentioned in Section IV-B1) as the data for learning, however, it was observed that the accuracy drops by 0.66%. Finally, the best result on the image based micro-gesture recognition's *Test* set outperforms the baseline by 26.67%.

B. Video Based Micro-Gesture Recognition

The results achieved on video dataset are discussed in detail. Table V shows the accuracy achieved on different experiments for videos. Here also, we used different cross-validation techniques and finally max voting is applied. Max voting is similar to that of the image-model. For using the temporal information, we perform experiments using LSTM and GRU. The results show that GRU performs better and the reason can be the presence of low sequence length due to which the GRU is able to learn the features more accurately. We achieve 67.50% accuracy on the *Test* set by applying GRU with 6-fold cross-validation. We also attempted to use augmented data for training but no improvement is observed. Our model performs surprisingly well when we tried to ignore the LPQ features, here 69.17% classification accuracy is observed. It seems that the complimentary information with the hand crafted features is not giving better scores on the video based task. This can be attributed to the pooling step. The baseline paper achieves the maximum accuracy of 66.70% by using LPQTOP features trained over SVM classifier. Finally, the method's best result for the video based micro-gesture recognition outperforms the baseline by 2.47%.

Table II: Confusion matrix of the image based challenge on the *Validation* set. Here the rows represent the predicted classes.

Accuracy=75.76%	Class 1	Class 2	Class 3	Total
Class 1	**2079**	275	46	2400
Class 2	185	**1229**	398	1812
Class 3	2	684	**1662**	2348
Total	2266	2188	2106	**6560**

Table III: Precision, recall and F1 score for three classes.

Accuracy=75.76%	Precision	Recall	F1 Score
Class 1	0.86	0.91	0.88
Class 2	0.67	0.56	0.61
Class 3	0.70	0.78	0.73

Table IV: Classification accuracy on the *Test* set (in %) for image based micro-gesture recognition.

Model	Training Data Set	*Test* Acc. (%)
Baseline paper [19]	*Train* only	50.90
4-fold validation, top-1	*Train* + *Validation*	73.52
4-fold validation, max voting	*Train* + *Validation*	77.06
3-fold validation, max voting	*Train* + *Validation*	**77.57**
4-fold validation, max voting	*Train* videos + *Train* + *Validation* images	76.91

VI. Conclusion and Future works

Efficient gesture recognition in Holoscopic images can mark the beginning of new era of 3D imaging technology. This paper presents a deep learning based approach for micro-gestures recognition in Holoscopic images and videos. The experiments shows an improvement over the state of the art methods. We explore the use of handcrafted as well

Table V: Classification accuracy (in %) on *Test* set for video based micro-gesture recognition.

Model	Training Data Set	*Test* Acc. (%)
Baseline paper [19]	*Train* only	66.70
6-fold validation, LSTM	*Train* + *Validation*	63.33
9-fold validation, LSTM	*Train* + *Validation*	65.41
6-fold validation, GRU	*Train* + *Validation*	67.50
9-fold validation, GRU	*Train* + *Validation*	65.83
4-fold validation, GRU	augmented *Train* + *Validation*	**67.50**
4-fold validation, GRU (without LPQ)	augmented *Train* + *Validation*	**69.17**

as the features extracted from deep learning methods and tried to explore the variation in the methods to improve the accuracy of gesture recognition. We find the features extracted from deep learning models to have better influence on the performance than handcrafted features.

To increase the applications of Holoscopic images, the work done in this paper can be extended for recognizing dynamic hand gestures. Recognizing dynamic hand gestures can be used to provide more information to the 3D objects. The accuracy can be further improved by using techniques to remove the noise while extracting VPI's and using them for training in CNN.

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